

Conductor-Shell p-Adic Fractional Heat Semigroup: Exact Zeta, Scale Oscillation, and Boundary Closure

Route-A source-local certificate HCS-C283

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Abstract

For every fixed prime p and $\alpha > 0$, we define a nonnegative operator on $L^2(\mathbb{Z}_p)$ directly as the Fourier multiplier 0 on constants and $p^{\alpha n}$ on characters of exact conductor n . Its spectrum and a conditional-expectation reconstruction prove a positive conservative Markov semigroup with an exact heat trace. We prove the exact eigenvalue staircase, a discrete-scale oscillation and log-periodic heat profile, and the sharp Schatten threshold for fractional resolvents. The mean-zero spectral zeta has a complete vertical pole lattice and an explicit primed determinant; all $\alpha = 0$, $\mu = 0$, and $t = 0$ faces and a finite-quotient DFT certificate are closed. The fixed local prime is not promoted to all-prime target arithmetic.

1 The frozen conductor multiplier

Let p be any fixed rational prime, let $G = \mathbb{Z}_p$ have Haar mass one, and use $\widehat{G} = \mathbb{Q}_p/\mathbb{Z}_p$. A nonzero class ξ has exact conductor $n(\xi) = n$ when χ_ξ is trivial on $p^n\mathbb{Z}_p$ but not on $p^{n-1}\mathbb{Z}_p$. Exactly $p^n - p^{n-1} = (p-1)p^{n-1}$ characters have conductor n .

For $\alpha > 0$ we freeze

$$D_{p,\alpha}1 = 0, \quad D_{p,\alpha}\chi_\xi = p^{\alpha n(\xi)}\chi_\xi, \quad (1)$$

with its maximal diagonal domain. For $\mu \geq 0$, set $A_{p,\alpha,\mu} = D_{p,\alpha} + \mu I$ and $T_t = e^{-tA_{p,\alpha,\mu}}$. Equation (1) is the normalization: the paper does not identify it with every convention called a Vladimirov operator. Pseudodifferential spectra over p-adic fields, p-adic wavelet eigenbases, and ultrametric Markov semigroups form the surrounding literature [1, 2, 3, 5]. More directly, Example 5.1 of Chacón-Cortés-Zúñiga-Galindo [4], at dimension one and exponent α , already gives exactly our positive shell spectrum, multiplicities, geometric zeta, and vertical pole lattice. Those receive zero priority credit. We retain a source-local Markov, scale-boundary, determinant, and finite-reconstruction closure, not a literature-novelty claim.

Let E_n denote conditional expectation onto functions constant on cosets of $p^n\mathbb{Z}_p$, so $E_0 = P_0$ projects onto constants and $E_n \uparrow I$ strongly. Put

$$W_0 = \text{ran } E_0, \quad W_n = \text{ran } E_n \ominus \text{ran } E_{n-1} \quad (n \geq 1).$$

Then $\dim W_n = (p-1)p^{n-1}$ and the character and filtration descriptions give the same orthogonal decomposition.

Theorem 1 (spectrum and Markov reconstruction). *The spectrum of $D_{p,\alpha}$ consists of the simple eigenvalue zero and $p^{\alpha n}$ with multiplicity $(p-1)p^{n-1}$, $n \geq 1$. Its resolvent is compact. The spectrum of $A_{p,\alpha,\mu}$ is the simple eigenvalue μ and $\mu + p^{\alpha n}$ with the same multiplicities. Moreover $e^{-tD_{p,\alpha}}$ is a positive, conservative, self-adjoint contraction semigroup, while T_t is positive sub-Markov and $\|T_t\| = e^{-\mu t}$.*

Proof. Character counting and (1) prove the spectral assertion; the eigenvalues tend to infinity. A second reconstruction is

$$D_{p,\alpha} = \sum_{n \geq 1} p^{\alpha n} P_{W_n} = \sum_{n \geq 0} a_n (I - E_n), \quad (2)$$

where $a_0 = p^\alpha$ and $a_n = p^{\alpha(n+1)} - p^{\alpha n}$ for $n \geq 1$. On W_m , the right side is $\sum_{n < m} a_n = p^{\alpha m}$. For every conditional expectation E ,

$$e^{-sa(I-E)} = E + e^{-sa}(I-E)$$

is positive, constant-preserving, and contractive. The finite factors in (2) commute; their strong limit proves the Markov claims. The scalar killing factor $e^{-\mu t}$ gives the final norm and sub-Markov statement. $\square$

This proof derives positivity from the residue filtration. Nonnegative eigenvalues alone would not have established positivity preservation.

2 Heat trace, exact staircase, and Schatten boundary

For $q > 0$, write $\mathcal{S}_q$ for the operators whose singular values have finite q -sum. When $0 < q < 1$, this is the quasi-Schatten ideal with its usual quasi-norm, not a Banach Schatten class.

Theorem 2 (sharp scale theorem). *For $t > 0$, $T_t \in \mathcal{S}_q$ for every $q > 0$ and*

$$\mathrm{Tr} T_t = e^{-\mu t} \left[1 + \sum_{n \geq 1} (p-1) p^{n-1} e^{-tp^{\alpha n}} \right]. \quad (3)$$

If $N(\Lambda)$ counts the positive spectrum, then

$$p^{\alpha m} \leq \Lambda < p^{\alpha(m+1)} \implies N(\Lambda) = p^m - 1. \quad (4)$$

Consequently

$$\limsup_{\Lambda \rightarrow \infty} \frac{N(\Lambda)}{\Lambda^{1/\alpha}} = 1, \quad \liminf_{\Lambda \rightarrow \infty} \frac{N(\Lambda)}{\Lambda^{1/\alpha}} = \frac{1}{p}.$$

For $\sigma, q > 0$,

$$(I + D_{p,\alpha})^{-\sigma} \in \mathcal{S}_q \iff \alpha \sigma q > 1; \quad (5)$$

the equality case diverges.

Proof. The spectral theorem and shell multiplicities give (3), and exponential decay in $p^{\alpha n}$ dominates every geometric multiplicity; the same argument applies after taking any positive power of the singular values. Summing the first m multiplicities gives $p^m - 1$, proving (4). At a shell and just before its successor, the scaled ratios approach 1 and $1/p$, respectively. Finally, the q -sum in (5) is comparable, up to finitely many terms, to $\sum_n p^{n(1-\alpha\sigma q)}$, which converges exactly under the displayed strict inequality. $\square$

The staircase also has an exact small-time scaling profile. If $t = p^{-\alpha m} \tau$, then locally uniformly for $\tau > 0$,

$$t^{1/\alpha} (\mathrm{Tr} e^{-tD_{p,\alpha}} - 1) \xrightarrow{m \rightarrow \infty} \Phi_{p,\alpha}(\tau) = \frac{p-1}{p} \tau^{1/\alpha} \sum_{j \in \mathbb{Z}} p^j e^{-\tau p^{\alpha j}}. \quad (6)$$

The negative tail is geometric and the positive tail superexponential, so the limit follows by reindexing $j = n - m$. Reindexing once more gives $\Phi_{p,\alpha}(p^\alpha \tau) = \Phi_{p,\alpha}(\tau)$. Equations (4) and (6) record a genuine discrete-scale oscillation, not a missing smooth Weyl constant. Notice the category distinction: positive-time heat is in every $\mathcal{S}_q$, whereas the fractional resolvent has the sharp boundary (5).

3 Complete zeta lattice and primed determinant

All zeta statements are on $L_0^2(\mathbb{Z}_p)$; the prime symbol means that the constant zero mode is omitted.

Theorem 3 (mean-zero spectral zeta). *Initially for $\Re s > 1/\alpha$,*

$$\zeta_{p,\alpha}(s) = \mathrm{Tr}' D_{p,\alpha}^{-s} = (1 - p^{-1}) \frac{p^{1-\alpha s}}{1 - p^{1-\alpha s}}. \quad (7)$$

This is the meromorphic continuation. Its complete pole set is

$$s_k = \frac{1}{\alpha} + \frac{2\pi i k}{\alpha \log p}, \quad k \in \mathbb{Z}, \quad (8)$$

all poles are simple, and each residue is $(1 - p^{-1})/(\alpha \log p)$. Furthermore

$$\zeta_{p,\alpha}(0) = -1, \quad \zeta'_{p,\alpha}(0) = -\frac{\alpha \log p}{p-1}, \quad \det' \zeta D_{p,\alpha} = p^{\alpha/(p-1)}. \quad (9)$$

Proof. Shell summation gives a geometric series of ratio $p^{1-\alpha s}$, proving (7). The denominator vanishes exactly at (8); its derivative there is $\alpha \log p$, while the numerator is $1 - p^{-1}$, proving completeness, simplicity, and the residue. Substitution and differentiation at zero give (9), including the convention $\det' \zeta D = \exp[-\zeta'(0)]$. $\square$

The vertical pole lattice is the Mellin counterpart of discrete spatial scaling. It is not a target zero set, and the source-local primed determinant is not a target Euler product.

4 Boundary atlas and finite-quotient reconstruction

At $\alpha = 0$ the natural face is $D_{p,0} = I - P_0$. It is bounded and noncompact, with eigenvalue one of infinite multiplicity. For fixed $t > 0$, $e^{-t(D_{p,\alpha} + \mu I)}$ converges strongly as $\alpha \downarrow 0$ to $e^{-t(I - P_0 + \mu I)}$, but not in operator norm: the mean-zero norm defect is $e^{-(\mu+1)t}$. Thus trace class, compact resolvent, and (7) do not extend to that face.

At $\mu = 0$ constants remain the zero mode and the heat semigroup is Markov; this is why (7)–(9) are mean-zero and primed. At $t = 0$, $T_0 = I$ is not compact and belongs to no finite $\mathcal{S}_q$. Conversely, for fixed $t > 0$, $\alpha \rightarrow \infty$ gives $T_t \rightarrow e^{-\mu t} P_0$ in operator norm, since the mean-zero norm is $e^{-\mu t - t p^\alpha}$. No odd-prime step was used, so $p = 2$ is included.

On $G_N = \mathbb{Z}/p^N \mathbb{Z}$, character $k \neq 0$ has conductor $N - v_p(k)$. Hence the finite DFT multiplier agrees exactly with

$$\sum_{n=1}^N p^{\alpha n} (E_n - E_{n-1}). \quad (10)$$

The executable receipt tests 36 DFT instances through quotient order 4096, alongside 402 analytic and control cells. A separate checker performs 1507 assertions, SymPy supplies 102 exact rational-matrix and zeta checks, fresh replay is byte-identical, and all 17 repaired-hash hostile mutations fail.

5 Route-A obstruction

The prime p is intrinsic to the local source, so the ceiling is honestly `A0_WEAK_ARITHMETIC_RELATION`. But one fixed p is not all rational primes; conductor shells are scale eigenspaces, not primitive orbits carrying logarithmic prime lengths. Composite branching numbers 4, 6, 10 reproduce the spectrum, geometric zeta, counting, and Schatten algebra, which is the strongest proves-too-much control. There is no target determinant, analytic divisor bridge, or same-clock arithmetic quantization.

Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the tuple is

(`A0_WEAK_ARITHMETIC_RELATION`, `A1_FAIL`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`),

overall `ROUTE_A_REJECTED`; Route B is disabled. No fixed local prime, source heat trace, or zeta determinant is promoted to target Euler data, bad-prime data, root numbers, or a Hilbert–Pólya operator.

References

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